

Regularization techniques:

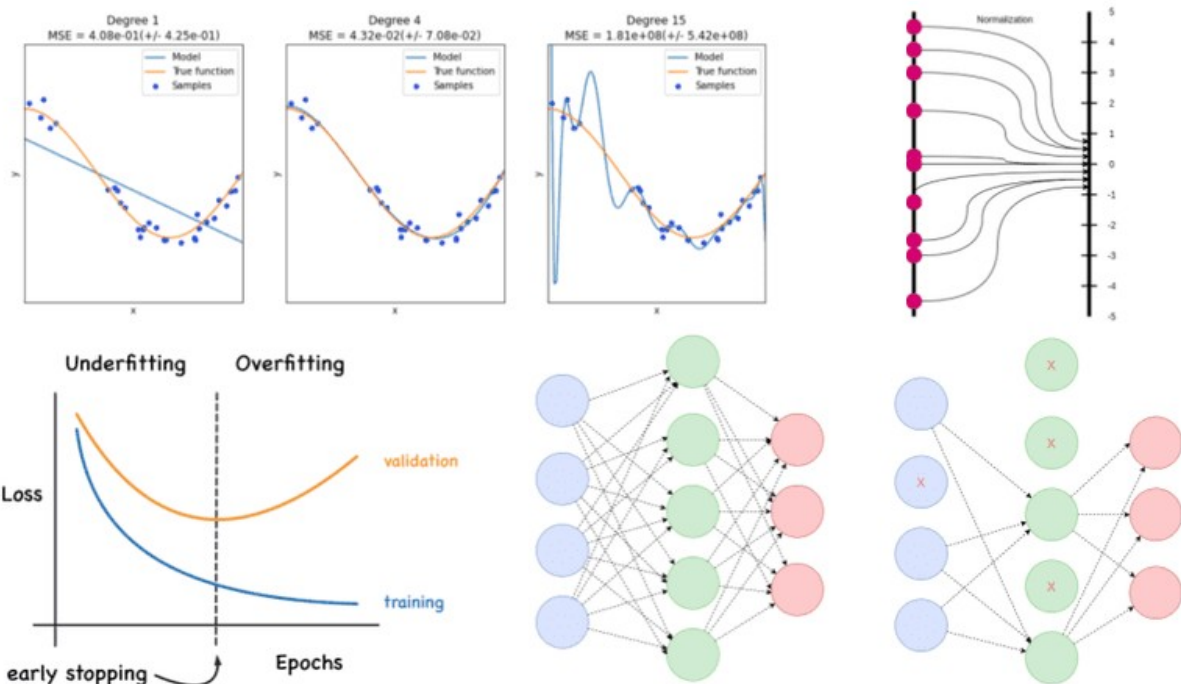
are ways to reduce the number of active parameters in order to ultimately overfitting

1. **L1 Regularization (Lasso):**
 - Adds the absolute values of the weights to the loss function, penalizing large weights. Leads to sparse weight vectors as it encourages some weights to be exactly zero.
2. **L2 Regularization (Ridge):**
 - Adds the squared values of the weights to the loss function, penalizing large weights. Encourages small weight values and smoother weight vectors.
3. **Elastic Net Regularization:**
 - A combination of L1 and L2 regularization, allowing you to control the balance between the two.
4. **Dropout:**
 - During training, randomly sets a fraction of the neurons' outputs to zero, effectively dropping them out. Prevents co-adaptation of neurons and encourages the network to be more robust.
5. **Batch Normalization:**
 - Normalizes the activations of each layer within a mini-batch to have zero mean and unit variance. Helps stabilize and speed up training by reducing internal covariate shift.
6. **Weight Decay:**
 - Adds a penalty term to the loss function that discourages large weight values. Equivalent to L2 regularization.
7. **Early Stopping:**
 - Monitors the validation performance during training and stops training when the performance starts to degrade, preventing the model from overfitting.
8. **Data Augmentation:**
 - Generates additional training data by applying various transformations to the existing data, such as rotation, scaling, flipping, and cropping. Increases the diversity of the training set and reduces overfitting.
9. **DropConnect:**
 - Similar to dropout but applies the dropout mask to the connections between neurons instead of the neuron outputs. Randomly sets weights to zero during training.
10. **Gradient Clipping:**
 - Limits the magnitude of gradients during training to prevent exploding gradient problems.
11. **Noise Injection:**
 - Adds noise to the inputs or intermediate layers during training. Acts as a form of regularization by making the network more robust to noise.
12. **Label Smoothing:**

- Instead of using hard binary labels, assigns a soft label distribution to each class. Reduces the model's confidence and encourages it to learn more generalizable features.

These techniques can be used individually or in combination, depending on the problem and the architecture of the neural network. The choice of regularization techniques often depends on the specific characteristics of the dataset and the complexity of the model.

- the most used ones are:
 - **L1 and L2**
 - **Dropout**
 - **Early stopping**
 - **Data augmentation**



- <https://theaisummer.com/regularization/>